

Remediation Scoring Model for Database Migrations

Overview

The **Remediation Score** is a quantitative model designed to estimate the effort required to address code incompatibilities or transformation challenges during a database or application migration. This model supports structured planning by categorizing remediation tasks into **Easy**, **Medium**, and **Hard** levels of complexity and assigning weights to each level.

Formula Definition

$$\text{Remediation Score} = (w_E * n_E) + (w_M * n_M) + (w_H * n_H)$$

Where:

- w_E : Weight assigned to an *Easy* remediation operation in hours
- w_M : Weight assigned to a *Medium* remediation operation in hours
- w_H : Weight assigned to a *Hard* remediation operation in hours
- n_E : Number of observed Easy-level items
- n_M : Number of observed Medium-level items
- n_H : Number of observed Hard-level items

Example Weights and Observations

Assume the following:

Complexity Level	Weight in hours	Observed Count
Easy	1	20
Medium	3	10
Hard	8	5

Then:

$$\text{Remediation Score} = (1 * 20) + (3 * 10) + (8 * 5) = 20 + 30 + 40 = 90 \text{ hours}$$

Normalized Remediation Score

To provide a comparable metric across projects or components, normalize the score:

Normalized Score = Remediation Score / (n_E + n_M + n_H)

In the above example:

Normalized Score = 90 / (20 + 10 + 5) = 90 / 35 ≈ 2.57

Benefits of Normalizing the Remediation Score

Normalization transforms the raw remediation score into a scale-independent metric, allowing for meaningful comparisons across codebases or modules of varying sizes.

Benefits:

1. Comparability

Normalized scores enable direct comparison between systems, regardless of the total number of operations.

- Example:

- Module A: Raw Score = 90, Total Ops = 30 → Normalized Score = 3.0
- Module B: Raw Score = 90, Total Ops = 90 → Normalized Score = 1.0

2. Effort Intensity Insight

Normalization answers “how complex is the average remediation task?” rather than just total effort.

A higher normalized score suggests higher per-item complexity.

3. Prioritization

Normalized scores support better prioritization when evaluating which code modules or features to remediate first, especially when resource constrained.

Purpose and Use Cases

- **Effort estimation:** Inform project timelines and resourcing needs
- **Risk identification:** Highlight areas of high transformation complexity
- **Prioritization:** Guide incremental migration phases or automation focus

Adaptability

- The weights (w_E, w_M, w_H) may be adjusted based on:
 - Estimated hours per remediation type (current approach)
 - Relative financial cost
 - Risk or business criticality
- Additional complexity categories may be added as needed

- Remediation tasks can be auto-generated by analysis tools and/or manually reviewed
-

Output and Interpretation

- A **higher Remediation Score** indicates greater expected remediation effort
 - The **normalized score** helps compare the complexity intensity across modules
 - Combined with cost/time conversion ratios, this score can be translated into budget or sprint allocations
-

Available Tools

A companion Excel spreadsheet implements an example of this model with:

- Editable weight and count inputs
 - Automatic calculation of scores
-

Notes

- Questions/comments, please contact the author [Matt DeMarco](mailto:matthew.demarco@oracle.com)